Q1.

A curve has polar equation $r = 3 + 2\cos\theta$, where $0 \le \theta \le 2\pi$

(a) (i) State the maximum and minimum values of r.

(2)

(ii) Sketch the curve.

(2)

(b) The curve $r = 3 + 2\cos\theta$ intersects the curve with polar equation $r = 8\cos^2\theta$, where $0 \le \theta \le 2\pi$

Find all of the points of intersection of the two curves in the form $[r, \theta]$.

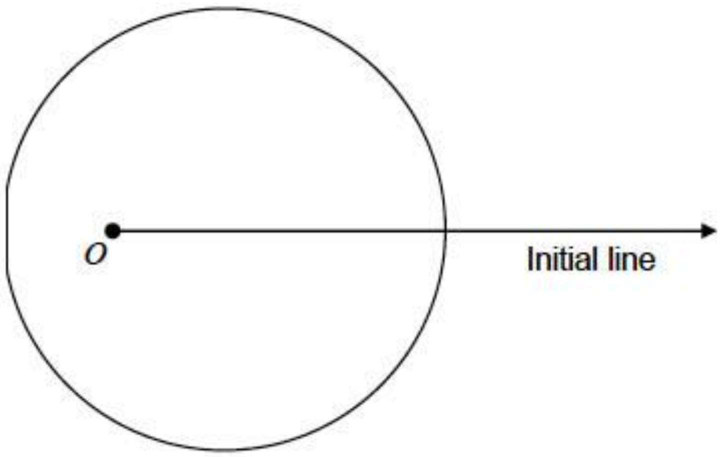
You should leave your answers in exact form where possible.

(6)

(Total 10 marks)

Q2.

The diagram shows a sketch of a curve C, the pole O and the initial line.



The polar equation of C is $r = 4 + 2\cos\theta$, $\quad -\pi \le \theta \le \pi$

(a) Show that the area of the region bounded by the curve C is 18π

(4)

(b) Points A and B lie on the curve C such that $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$ and AOB is an equilateral triangle.

Find the polar equation of the line segment AB

(4)

(Total 8 marks)

Q3.

The Cartesian equation of a circle is $(x + 8)^2 + (y - 6)^2 = 100$.

Using the origin O as the pole and the positive x-axis as the initial line, find the polar equation of this circle, giving your answer in the form $r = p \sin\theta + q \cos\theta$.

(Total 4 marks)

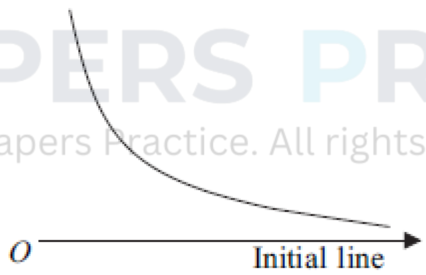
Q4.

(a) A curve has cartesian equation $xy = 8$. Show that the polar equation of the curve is $r^2 = 16 \operatorname{cosec} 2\theta$.

(3)

(b) The diagram shows a sketch of the curve, C, whose polar equation is

$$r^2 = 16 \operatorname{cosec} 2\theta, \qquad 0 < \theta < \frac{\pi}{2}$$



(i) Find the polar coordinates of the point N which lies on the curve C and is closest to the pole O.

(2)

(ii) The circle whose polar equation is $r = 4\sqrt{2}$ intersects the curve C at the points P and Q. Find, in an exact form, the polar coordinates of P and Q.

(4)

(iii) The obtuse angle PNQ is a radians. Find the value of α, *giving your answer to three significant figures.*

(5)

(Total 14 marks)

Q5.

A curve C has polar equation $r(1 + \cos\theta) = 2$.

(a) Find the cartesian equation of C, giving your answer in the form $y^2 = \mathrm{f}(x)$.

(5)

(b) The straight line with polar equation $4r = 3\sec\theta$ intersects the curve C at the points P and Q. Find the length of PQ.

(4)

(Total 9 marks)

Q6.

The curve C_1 is defined by $r = 2\sin\theta$, $0 \le \theta < \frac{\pi}{2}$.

The curve C_2 is defined by $r = \tan\theta$, $0 \le \theta < \frac{\pi}{2}$.

(a) Find a cartesian equation of C_1

(3)

(b) (i) Prove that the curves C_1 and C_2 meet at the pole O and at one other point, P, in the given domain. State the polar coordinates of P.

(4)

(ii) The point A is the point on the curve C_1 at which $\theta = \frac{\pi}{4}$.

The point B is the point on the curve C_2 at which $\theta = \frac{\pi}{4}$.

Determine which of the points A or B is further away from the pole O, justifying your answer.

(2)

(iii) Show that the area of the region bounded by the arc OP of C_1 and the arc OP of C_2 is $a\pi + b\sqrt{3}$, where a and b are rational numbers.

(10)

(Total 19 marks)

Q7.

The polar equation of a curve C_1 is

$$r = 2(\cos\theta - \sin\theta),\ 0 \le \theta \le 2\pi$$

(a) (i) Find the cartesian equation of C_1.

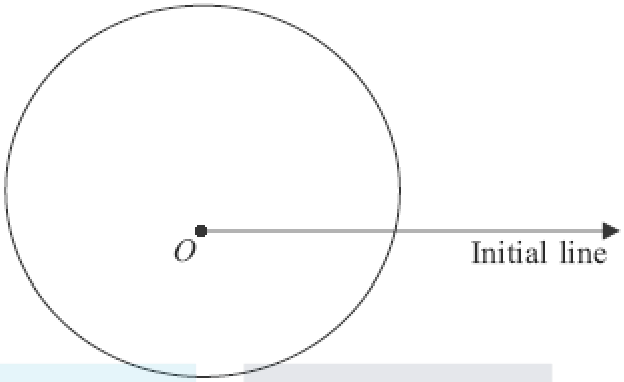
(4)

(ii) Deduce that C_1 is a circle and find its radius and the cartesian coordinates of its centre.

(3)

(b) The diagram shows the curve C_2 with polar equation

$$r = 4 + \sin\theta, \quad 0 \le \theta \le 2\pi$$



(i) Find the area of the region that is bounded by C_2.

(6)

(ii) Prove that the curves C_1 and C_2 do not intersect.

(4)

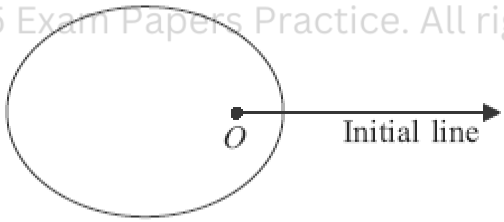
(iii) Find the area of the region that is outside C_1 but inside C_2.

(2)

(Total 19 marks)

Q8.

The diagram shows a sketch of a curve C, the pole O and the initial line.



The curve C has polar equation

$$r = \frac{2}{3 + 2\cos\theta}, \quad 0 \le \theta \le 2\pi$$

(a) Verify that the point L with polar coordinates $(2, \pi)$ lies on C.

(1)

(b) The circle with polar equation $r = 1$ intersects C at the points M and N.

(i) Find the polar coordinates of M and N.

(3)

(ii) Find the area of triangle LMN.

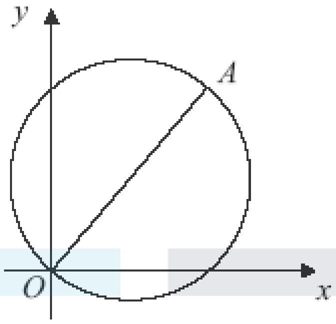
(4)

(c) Find a cartesian equation of C, giving your answer in the form $9y^2 = f(x)$.

(5)
(Total 13 marks)

Q9.

The diagram shows a sketch of a circle which passes through the origin O.



The equation of the circle is $(x - 3)^2 + (y - 4)^2 = 25$ and OA is a diameter.

(a) Find the cartesian coordinates of the point A.

(2)

(b) Using O as the pole and the positive x-axis as the initial line, the polar coordinates of A are (k, α).

(i) Find the value of k and the value of $\tan \alpha$.

(2)

(ii) Find the polar equation of the circle $(x - 3)^2 + (y - 4)^2 = 25$, giving your answer in the form $r = p \cos \theta + q \sin \theta$.

(4)
(Total 8 marks)

Q10.

A curve C has polar equation

$$r^2 \sin 2\theta = 8$$

(a) Find the cartesian equation of C in the form $y = f(x)$.

(3)

(b) Sketch the curve C.

(1)

(c) The line with polar equation $r = 2 \sec \theta$ intersects C at the point A. Find the polar coordinates of A.

(4)

(Total 8 marks)

Q11.

(a) Show that $x^2 = 1 - 2y$ can be written in the form $x^2 + y^2 = (1 - y)^2$.

(1)

(b) A curve has cartesian equation $x^2 = 1 - 2y$.

Find its polar equation in the form $r = f(\theta)$, given that $r > 0$.

(5)
(Total 6 marks)

Q12.

A curve has polar equation $r(1 - \sin\theta) = 4$. Find its cartesian equation in the form $y = f(x)$.

(Total 6 marks)

Q13.

(a) Show that $(\cos\theta + \sin\theta)^2 = 1 + \sin 2\theta$.

(1)

(b) A curve has cartesian equation

$(x^2 + y^2)^3 = (x + y)^4$

Given that $r \geq 0$, show that the polar equation of the curve is

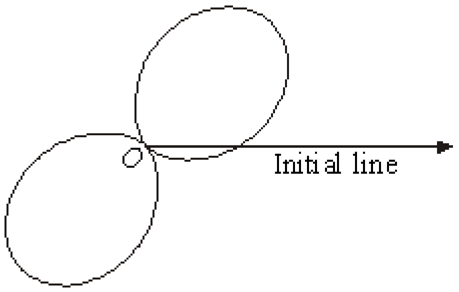
$r = 1 + \sin 2\theta$

(4)

(c) The curve with polar equation

$r = 1 + \sin 2\theta, \quad -\pi \leq \theta \leq \pi$

is shown in the diagram.



(i) Find the two values of θ for which $r = 0$.

(3)

(ii) Find the area of one of the loops.

(6)

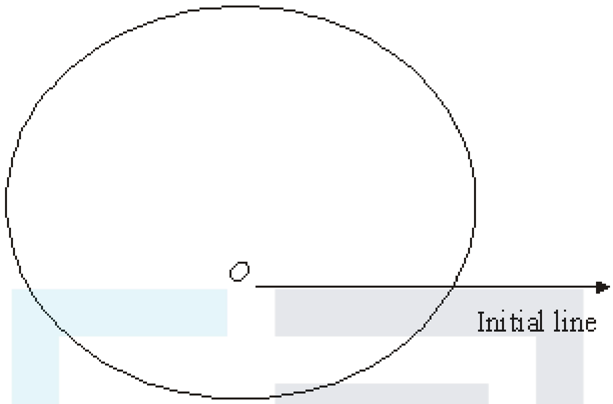
(Total 14 marks)

Q14.

(a) A circle C_1 has cartesian equation $x^2 + (y - 6)^2 = 36$. Show that the polar equation of C_1 is $r = 12 \sin \theta$.

(4)

(b) A curve C_2 with polar equation $r = 2\sin \theta + 5$, $0 \le \theta \le 2\pi$ is shown in the diagram.



Calculate the area bounded by C_2.

(6)

(c) The circle C_1 intersects the curve C_2 at the points P and Q. Find, in surd form, the area of the quadrilateral $OPMQ$, where M is the centre of the circle and O is the pole.

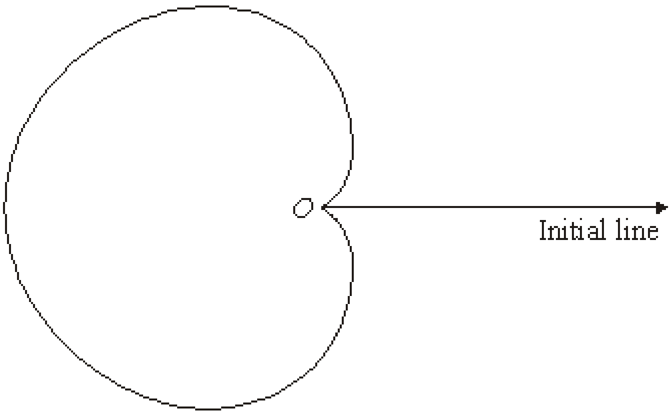
(6)

(Total 16 marks)

Q15.

The diagram shows the curve C with polar equation

$$r = 6(1 - \cos \theta), \qquad 0 \le \theta < \pi$$



(a) Find the area of the region bounded by the curve C.

(6)

(b) The circle with cartesian equation $x^2 + y^2 = 9$ intersects the curve C at the points A and B.

(i) Find the polar coordinates of A and B.

(4)

(ii) Find, in surd form, the length of AB.

(2)

(Total 12 marks)